35. SYNTHESIS OF THE GENITAL SYSTEM

DURATION : 03:26 STUDY SHEET

COURSE SUMMARY

This video offers an in-depth synthesis of genital development, highlighting embryological analogies and the differentiation of male and female sexual organs. You will learn how undifferentiated structures transform, under hormonal influence, to form distinct organs such as the penis and clitoris, while integrating the processes of midline closure. The concepts of gonadal sex, embryonic ducts, and somatic implications of psychic conflicts are also addressed, allowing for an enriching therapeutic approach for practitioners.

SYNTHESIS OF THE GENITAL SYSTEM

The development of the **genital system** shows remarkable correspondences between male and female structures, originating from the same undifferentiated embryo. These structures evolve under hormonal influence, modulating their progressive closure along the **midline**.

DIFFERENTIATION OF EXTERNAL GENITAL ORGANS

Initially, the structures are undifferentiated and then specialise into **male** or **female sexual organs**.

- **Genital tubercle:**

- Male: Glans and shaft of the penis

- Female: Glans and body of the clitoris

- **Urogenital sinus:**

- Male: Penile urethra

- Female: Vestibule of the vagina

- **Urethral fold:**

- Male: Penis and penile urethra

- Female: Labia minora

- **Labioscrotal fold:**

- Male: Scrotum

- Female: Labia majora

The progressive closure of these structures along the midline is a key process. This begins with the separation between the **cloaca** and the **rectum**.

The differentiation of the external genital organs is clearly visible:

- The clitoris develops into the labia minora and majora.
- In males, the embryonic clitoris becomes the glans and penis.
- The embryonic labia minora close to form the scrotum.

This differentiation begins between the **fifth** and **eighth week** of pregnancy, concurrently with hepatic and adrenal development.

MIDLINE AND EMBRYOLOGICAL ANALOGIES

A posterior view of the embryo reveals the **falciform ligament**, the **teres ligament**, and the **median fold**. The linea alba is the continuity of this tissue, resulting from closure along the midline. This is the culmination of all this fusion movement.

There is an analogy between the formation of the genital organs and that of other median structures of the body:

- The nose develops like a uterus.
- The sternum develops like a uterine system.

All these systems converge towards the midline, whether it is the sternum, the face, or the genital organs. This is the closure of the anterior midline.

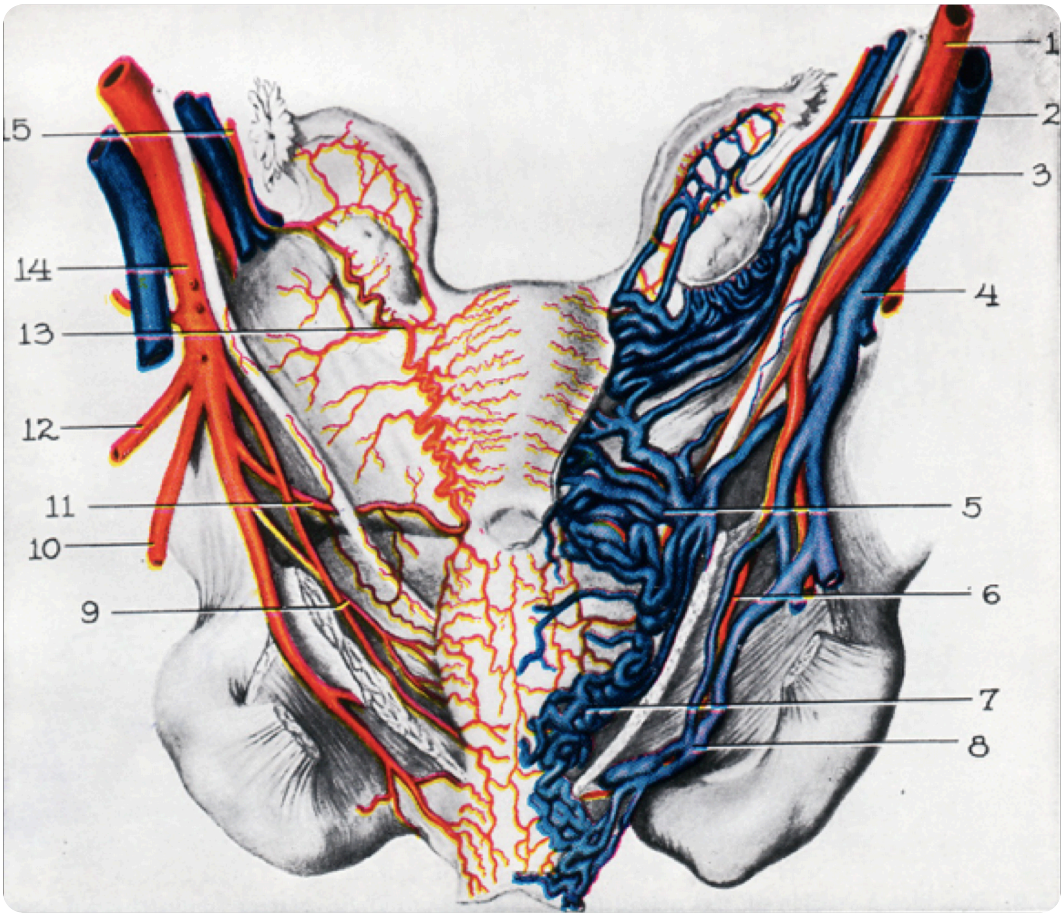

FIG. — Vascularization of cleft lip/palate

GONADAL SEX AND EMBRYONIC DUCTS

The development of the **genital ducts** depends on gonadal sex:

- **In the absence of androgen secretion (female):**

- The Wolffian duct regresses.

- The Müllerian duct develops to form the fallopian tube, uterus, and vagina.

- The urogenital sinus remains open (vestibule, labia minora). These structures, derived from the endoderm, will give rise to the urethra and the surrounding prostatic part.

- **In the presence of androgen secretion (male):**

- The Wolffian duct becomes the epididymis, vas deferens, seminal vesicle, and ejaculatory duct.

- The Müllerian duct regresses.

- The urogenital sinus closes.

The genital tubercle in males becomes the penis, and in females, the clitoris. The genital swellings form the labia minora and majora in females.

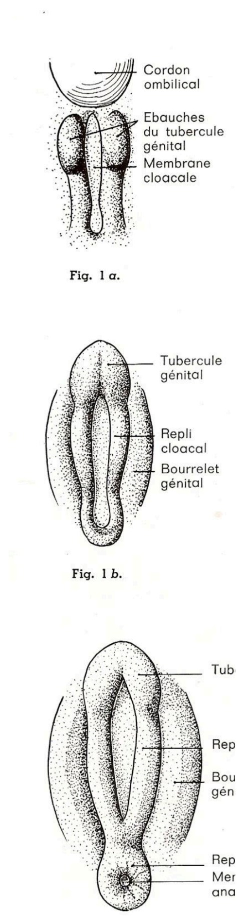

FIG. — Development of genital organs

SOMATISATION OF PSYCHIC CONFLICTS

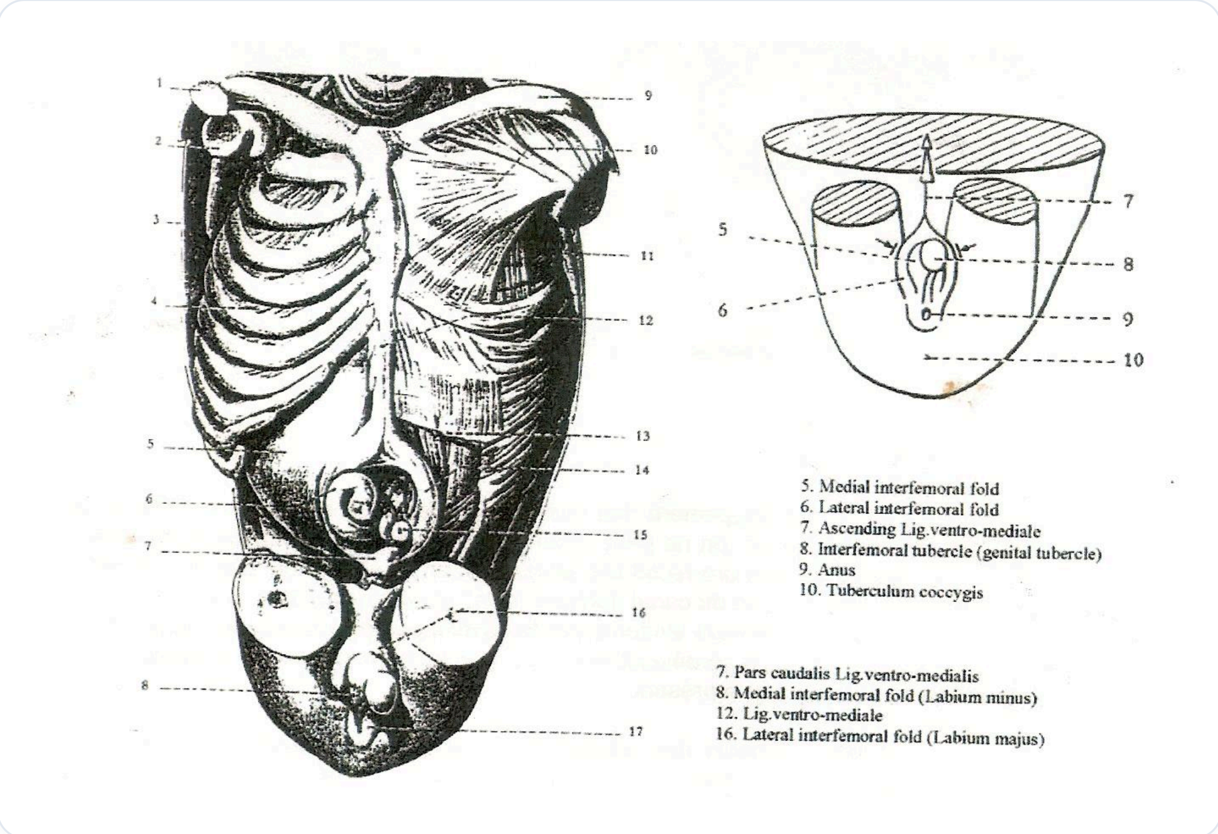

FIG. — External embryonic development

It is important to note that our body **somatises** psychic conflicts. If a conflict is not resolved, it can manifest physically according to its symbolism:

- **Separation:** Can affect the skin.
- **Urogenital, metabolic, rhythmic, or neurosensory conflict:** Will be expressed in the corresponding system.

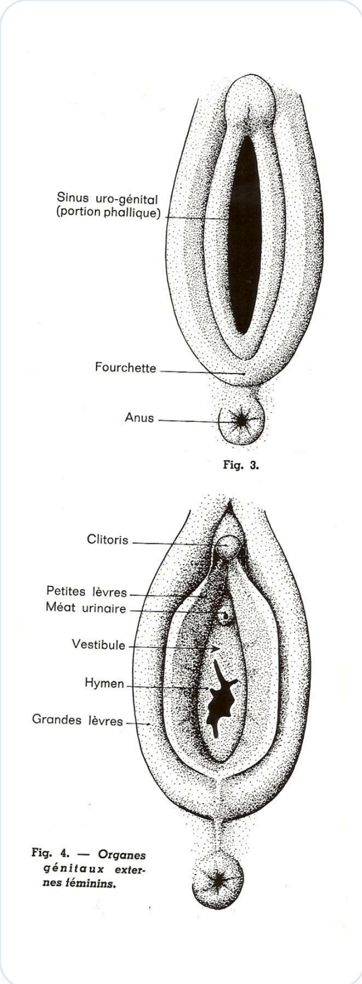

FIG. — Female genital organs

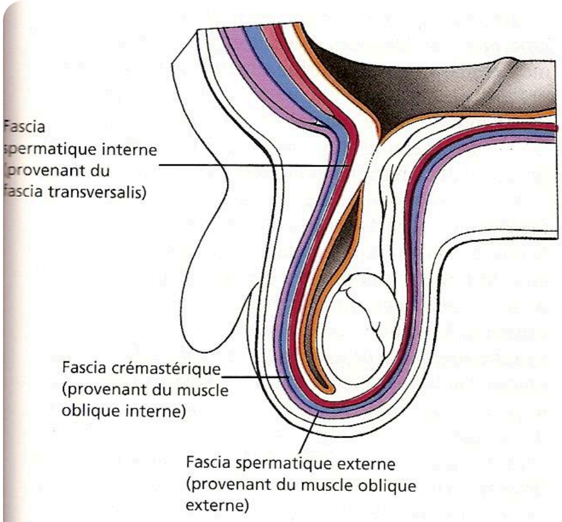

Fig. 10 - 21. Les trois feuillets évaginés de la paroi abdominale sont poussés dans le scrotum par le développement du processus vaginal et ils deviennent les trois enveloppes du cordon spermatique . Ces trois feuillets isolent le testicule et la vaginale dans un compartiment commun.

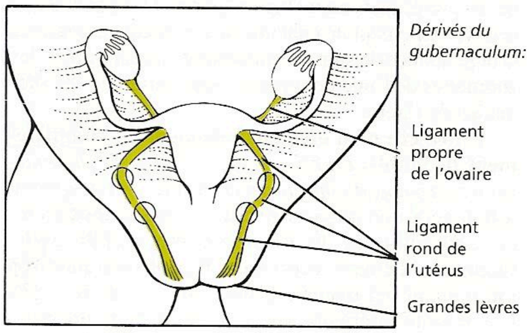

Fig. 10 - 22. Les ovaires descendent quelque peu au cours du développement et ils sont poussés vers les ligaments larges cependant que les conduits paramésonéphrotiques s'unissent pour former l'utérus. Dans le sexe féminin, le gubernaculum grandit à l'allure de celle du corps et il est attaché au conduit paramésonéphrotique au point de croisement avec la paroi postérieure du tronc. En conséquence, dans le sexe féminin, le vestige du gubernaculum unit les grandes lèvres de la vulve à la paroi de l'utérus et il est ensuite rabattu latéralement pour attacher l'ovaire.

FIG. — Descent of genital organs

Explanatory diagram

FIG. — Explanatory diagram

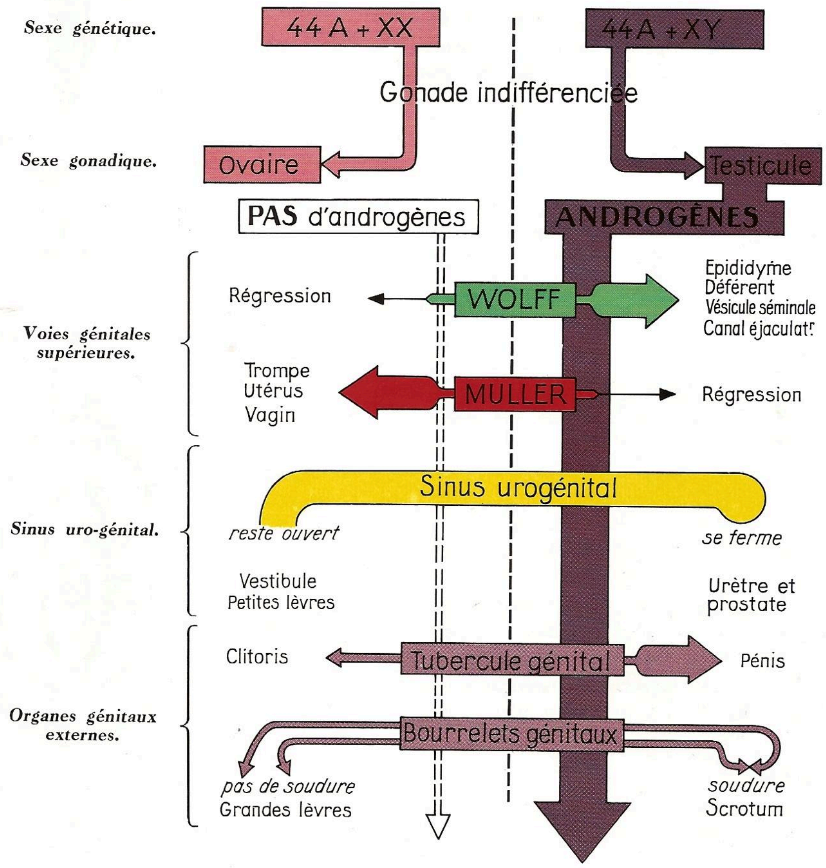

FIG. — Human sexual differentiation

Tableau 10 - 1. Développement des organes génitaux externes masculins et féminins

EBAUCHE PRESOMPTIVE	STRUCTURE MALE	STRUCTURE FEMELLE
Tubercule génital	Gland et corps du pénis	Gland et corps du clitoris
Sinus uro-génital définitif	Urètre pénien	Vestibule du vagin
Pli urétral	Pénis entourant l'urètre pénien	Petites lèvres de la vulve
Pli labio-scrotal	Scrotum	Grandes lèvres de la vulve

FIG. — External genital development

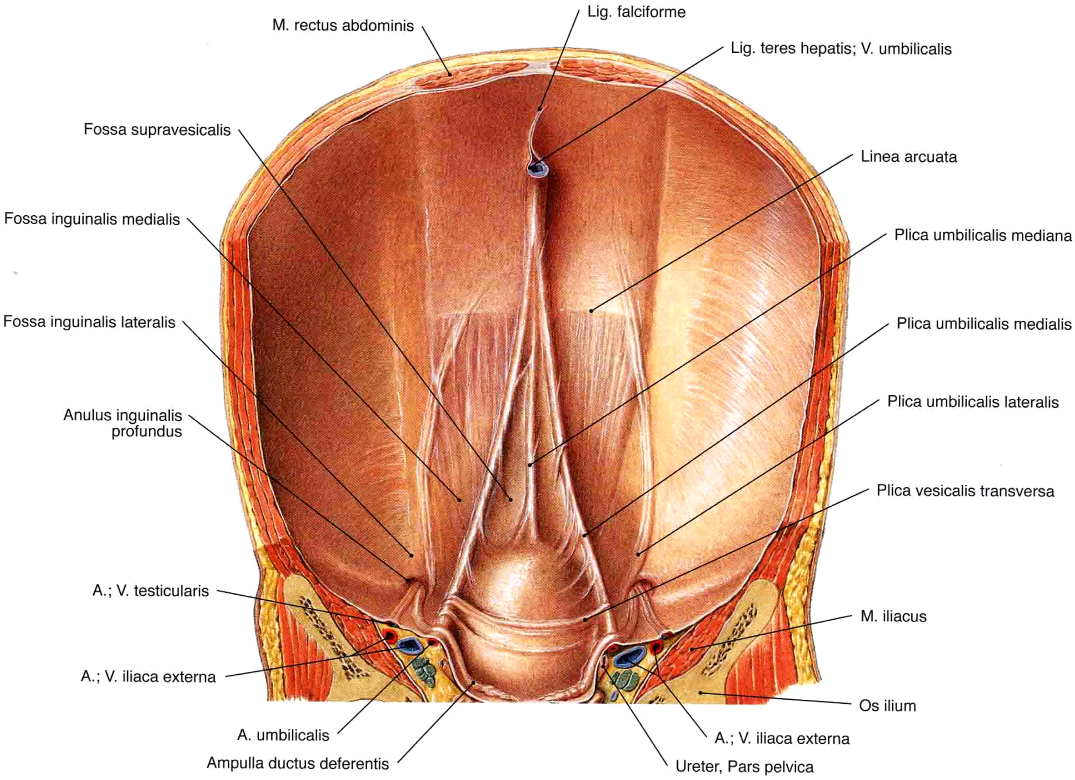

FIG. — Anterior abdominal wall